Samuele Pio Provenzano

Cheminformatics Engineer · Scientific Software Developer

📧 samupropio1@gmail.com · 📱 [phone] · 📍 [city, country] · 🔗 [LinkedIn] 🌐 Demo: https://samupropio1-ship-it.github.io/BioSpecInfo-v11/ · 💻 github.com/samupropio1-ship-it/BioSpecInfo-v11

> _Fields in square brackets `[…]` are to be completed before sending._

Profile

Chemist with a solid background in organic chemistry, spectroscopy and physical chemistry, combined with the ability to **design and build scientific software** using modern web technologies (WebAssembly, WebGL, PWA). I independently designed and developed **BioSpecInfo**, a client-side cheminformatics platform that runs molecular analysis and spectral prediction in the browser, offline and at zero server cost. I am seeking a role at the intersection of **chemistry and software development** (cheminformatics, molecular modelling, digital health, scientific software).

Core skills

Chemistry: organic chemistry and reaction mechanisms · NMR/IR/MS spectroscopy · stereochemistry (CIP) · retrosynthesis and synthons · physical chemistry and statistical thermodynamics · cheminformatics (descriptors, fingerprints, drug-likeness/ADMET).

Software: JavaScript (ES6/ESNext) · HTML5 · **WebAssembly** (RDKit, sql.js) · **WebGL** (three.js, 3Dmol.js) · SVG/Canvas 2D · **PWA**/Service Worker · Git/GitHub · E2E testing (Playwright) · REST API integration (PubChem).

Transferable: scientific rigour, autonomy, problem solving, technical documentation (SAD, V&V, compliance).

Flagship project — BioSpecInfo (PWA)

Open-access web platform for computational chemistry — sole design, development and validation.

- **Local-first architecture:** installable PWA of ≈4.5 MB, 100% offline, no

backend and no server cost.

- **Cheminformatics engine** RDKit (WebAssembly): SMILES/InChI/InChIKey/SMARTS,

descriptors, Morgan ECFP4 fingerprint + Tanimoto similarity, drug-likeness (Lipinski/Veber/Egan/Ghose/Muegge), Brenk/PAINS alerts.

- **Spectra Centre:** prediction of IR, $^{1}H/^{13}C$-NMR, MS with isotope pattern and

adduct masses, UV-Vis, Raman; functional-group highlighting on 2D and 3D structures.

- **Synthesis/Retrosynthesis:** 46 strategies with an animated mechanism player (2D/3D) showing electron pushing, intermediates and synthon formation.
- **Educational content:** 25 organic chemistry modules, 3D drug atlas (63 diseases), 3D astrochemistry, spaced-repetition quizzes (SM-2).
- **Quantitative validation** (thesis): ^{1}H-NMR predictor vs SDBS **MAE 0.31 ppm**; IR (C=O band) vs NIST **MAE 11 cm^{-1}**; **30–60 fps** from iPhone to Raspberry Pi 4.
- **Theoretical grounding:** SMILES notation treated as a generative formal language (molecular linguistics, Longo, *Symmetry* 2023).

Full technical documentation (architecture, validation, security, licensing) available in the repository (`docs/`).

Education

BSc/MSc in Chemistry — University of Bari Aldo Moro · [grade] · [month/year] Thesis: *"BioSpecInfo: an open-access web platform for computational chemistry and interactive consultation of chemical data on any device"* — Advisor: **Prof. Savino Longo**.

[Other qualifications, if any]

Languages

- Italian — native
- English — [level, e.g. B2/C1] (technical documentation written in English)

Links

- Live demo: https://samupropio1-ship-it.github.io/BioSpecInfo-v11/
- Code & documentation: github.com/samupropio1-ship-it/BioSpecInfo-v11
